LESSON PLAN

HIGH SPEED FLIGHT (12)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	02
Total No. of Lesson Plans	6	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Definition of terms, power absorption, feathering & braking & its advantages.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Definition of terms, power absorption, feathering & braking & its advantages.
5. Specific Objectives	12.1.1.1 Trainee will be able to describe the various types of propellers, principles of operation and pitch change mechanisms.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of propeller pitch change mechanisms diagrams; Safety: Ensure no loose papers; Supervision: Instructor monitors diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a propeller?
(b) Define the term 'blade angle'.
(c) List two types of propellers based on pitch control.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of terms • Power absorption concepts • Feathering mechanism • Braking mechanism • Advantages of feathering	L & Ds	PPT, IB	15 min
(b)	• Power absorption calculation • Feathering operational procedure • Braking operational procedure • Comparison of feathering benefits • Impact on engine safety	L & Ds	PPT, PDF	20 min
(c)	• Types of propellers • Principles of operation • Pitch change mechanisms • Application of braking systems • Summary of advantages	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how feathering reduces drag during engine failure.
(b) Describe the relationship between power absorption and propeller speed.
(c) Differentiate between static and dynamic braking in propeller systems.

12. Home Assignment and Follow Up	Prepare a brief note on how the knowledge of power absorption and feathering will assist in understanding the next lesson on co-axial and counter-rotating propellers.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Co-axial and counter rotating propellers, forces acting on propeller. Constant Speed Unit (CSU)	Refer to the master syllabus reference list	LP 03

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min